

Unit 7 Test 2 Review — ANSWER KEY (On-Level)**Part 1: Roots and Factors**

1. **A** ($x - 1$ is factor), **D** ($x = -2$ is zero)
(Note: C and E are also mathematically true statements.)

2.

- Factors: $(x + 2)$, $(x - 3)$, $(x + 3)$
- Not a Factor: $(x - 9)$

3.

- a) $(x - 1)$ is a factor.
- b) Factored: $(x - 1)(x - 5)(x + 2)$
- c) Zeros: $1, 5, -2$

Part 2: Factoring by Grouping

4. $(x^2 + 4)(x + 3)$

5. $(x - 3)(x + 3)(2x - 5)$

6. $(2x^2 - 5)(2x + 3)$

7. **B** ($x - 6$), **C** ($x^2 + 3$)

Part 3: Sum & Difference of Cubes

8. Warm-up:

- | | | |
|--------------|-------------|-------------|
| a) 2 | b) 5 | c) x |
| d) 3x | e) 4 | f) 6 |

9. **A** ($2x - 3$) and **C** ($4x^2 + 6x + 9$)

10. Linear: $x + 6$ Quadratic: $x^2 - 6x + 36$

11. $(5x - 2)(25x^2 + 10x + 4)$

12. $(3 + 4x)(9 - 12x + 16x^2)$

Part 4: Mixed Practice

13. Strategy: Grouping Answer: $(x - 3)(x + 3)(x + 4)$

14. Strategy: Diff of Cubes Answer: $(7x - 1)(49x^2 + 7x + 1)$

15. Strategy: Grouping Answer: $(3x^2 + 5)(x - 2)$

16. **FALSE.**

Explanation: If $f(2) = 0$, the zero is 2, so the factor is $(x - 2)$.